

A03 Neural Network Zoo

Reflection Journal

Working on the Neural Network Zoo project helped our group understand Convolutional Neural Networks in a much easier way. We used an **eagle to represent a CNN**, and that comparison made sense because eagles are known for strong vision and noticing small details. A CNN works in a similar way because it looks at images and learns important details such as edges, shapes, colors, textures, and patterns. Before this project, CNNs could seem confusing because there are many different parts and terms to remember, but using the eagle made it easier to picture what the network is actually doing.

One of the main things we learned is that a CNN does not look at a whole image and instantly knows what it is. It learns the image in steps. The first layers may notice simple things like lines, edges, curves, and colors. As the image moves through deeper layers, the CNN begins to notice more detailed features. For example, if it is looking at an eagle, it may start recognizing the **wings, feathers, beaks, head, and body shape**. All of those features are used together to help the CNN make a final prediction. This helped our group understand why CNNs have multiple layers and how each layer helps build a better understanding of the image.

The project also helped us understand ideas like **convolution, filters, feature maps, pooling, and flattening**. Convolution uses small filters that move across an image and look for important patterns. A filter may help the CNN notice an edge, line, or shape. Feature maps show where those patterns were found. Pooling then helps reduce the amount of information while keeping the most useful parts. This means the CNN does not need to keep every single pixel from the image. Flattening happens later, when the important image information is turned into a long list of values, so the network can use it to make a prediction. These terms sounded complicated at first, but seeing how they worked together made them much easier to understand.

We also learned more about how CNN improves while it is training. The model does not start out knowing the correct answers. It makes predictions, checks whether those predictions are right or wrong, and then adjusts itself. **Backpropagation** helps the model learn from its mistakes by changing the weights inside the network. After seeing many training images, the CNN can become more accurate. This connected well with what we had already learned about neural networks because it showed us that training is a repeated process of making a prediction, checking the error, and improving.

Another important part of the project was seeing how CNNs are used in the real world. CNNs can be used for **object detection, facial recognition, medical imaging, security systems, and self-driving cars**. For example, a self-driving car can use a computer vision to recognize other cars, people, road signs, and objects around it. CNNs can also be trained to find patterns in medical images. These examples helped show that CNNs are not just something we learn about in class. They are used in real systems and technology that people use every day.

Overall, the Neural Network Zoo project gave our group a better understanding of how CNNs work and why they are useful. The eagle example helped connect strong vision with the way a CNN studies an image and looks for important details. We now understand that a CNN learns simple features first, then builds those features into more detailed patterns, and finally uses what it learned to make a prediction. The project also helped us understand that different neural networks are designed for different types of problems. CNNs are especially useful for images because they are made to recognize visual patterns. After completing this project, terms like convolution, pooling, flattening, weights, and backpropagation became much easier to understand because we could connect them to the actual process instead of only trying to memorize definitions.